刘杨元琛

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科研与教学经历

博士后研究员 | 2024.10–至今 约翰霍普金斯大学 (Johns Hopkins University), 巴尔的摩, 美国

- 合作导师：Somnath Ghosh
- 结合科学机器学习（神经算子）与多尺度多物理场模型，开展无损评估（NDE）研究。

研究助理 | 2023.05–2023.08 阿贡国家实验室 (Argonne National Lab), 莱蒙特, 美国

- 面向高性能计算（HPC）的 MOOSE 有限元软件开发。

助教 | 2020.08–2021.05 昆山杜克大学, 昆山, 中国

- 课程：线性代数、随机数学；承担作业批改与习题课教学。

教育背景

博士 (PhD) | 2020.09–2024.09 杜克大学 (Duke University), 达勒姆, 美国

- 专业：机械工程（计算力学方向），导师：John Dolbow
- 将显式纳入材料强度面的相场断裂理论拓展至动态载荷，开展裂纹萌生与扩展的大规模模拟。
- 开展声学–弹性动力学–损伤耦合及脉冲激光烧蚀的建模与数值模拟。

硕士 (MS) | 2017.09–2020.03 上海交通大学, 上海, 中国

- 专业: 机械工程, 导师: 沈泳星

学士 (BE) | 2013.09–2017.07 吉林大学, 长春, 中国

- 专业：材料科学与工程

发表论文与预印本

[1] **Liu, Y.**, & Ghosh, S. (2026). Neural operator-based damage sensing digital twin for piezocomposites with parametrically upscaled coupled constitutive damage model. *Computer Methods in Applied Mechanics and Engineering*, *461*, 119169. https://doi.org/10.1016/j.cma.2026.119169

[2] Torres, D. E., Zeng, B., Costa, A., **Liu, Y.**, Jiang, W., Dolbow, J. E., & Hu, T. (2026). RACCOON: A versatile framework for variational approaches to fracture with coupled field phenomena. *Advances in Engineering Software*, *221*, 104242. https://doi.org/10.1016/j.advengsoft.2026.104242

[3] **Liu, Y.** (2024). A computational framework for simulating crack nucleation and growth in materials subjected to dynamic loads [PhD thesis]. In *Duke University* (p. 153).

[4] **Liu, Y.**, Lopez-Pamies, O., & Dolbow, J. E. (2024). On the effects of material strength in dynamic fracture: A phase-field study. *arXiv Preprint*. https://doi.org/10.48550/arXiv.2411.16393

[5] **Liu, Y.**, Zhong, P., Lopez-Pamies, O., & Dolbow, J. E. (2024). A model-based simulation framework for coupled acoustics, elastodynamics, and damage with application to nano-pulse lithotripsy. *International Journal of Solids and Structures*, *289*, 112626. https://doi.org/10.1016/j.ijsolstr.2023.112626

[6] Kumar, A., **Liu, Y.**, Dolbow, J. E., & Lopez-Pamies, O. (2024). The strength of the Brazilian fracture test. *Journal of the Mechanics and Physics of Solids*, *182*, 105473. https://doi.org/10.1016/j.jmps.2023.105473

[7] **Liu, Y.**, Claus, S., Kerfriden, P., Chen, J., Zhong, P., & Dolbow, J. E. (2023). Model-based simulations of pulsed laser ablation using an embedded finite element method. *International Journal of Heat and Mass Transfer*, *204*, 123843. https://doi.org/10.1016/j.ijheatmasstransfer.2022.123843

[8] Xiang, G., Chen, J., Ho, D., Sankin, G., Zhao, X., **Liu, Y.**, Wang, K., Dolbow, J., Yao, J., & Zhong, P. (2023). Shock waves generated by toroidal bubble collapse are imperative for kidney stone dusting during Holmium:YAG laser lithotripsy. *Ultrasonics Sonochemistry*, *101*, 106649. https://doi.org/10.1016/j.ultsonch.2023.106649

[9] **Liu, Y.**, Cheng, C., Ziaei-Rad, V., & Shen, Y. (2021). A micromechanics-informed phase field model for brittle fracture accounting for unilateral constraint. *Engineering Fracture Mechanics*, *241*, 107358. https://doi.org/10.1016/j.engfracmech.2020.107358

[10] Chen, C., **Liu, Y.**, He, X., Li, H., Chen, Y., Wei, Y., Zhao, Y., Ma, Y., Chen, Z., Zheng, X., & Liu, H. (2021). Multiresponse shape-memory nanocomposite with a reversible cycle for powerful artificial muscles. *Chemistry of Materials*, *33*(3), 987–997. https://doi.org/10.1021/acs.chemmater.0c04170

[11] **Liu, Y.**, Weng, K., & Shen, Y. (2020). A manifold learning approach to accelerate phase field fracture simulations in the representative volume element. *SN Applied Sciences*, *2*(10), 1682. https://doi.org/10.1007/s42452-020-03468-6

学术报告

2026 | USACM Student Chapter Seminar Series | Online | Seminar presentation

Damage sensing digital twin for piezocomposites integrating neural operator with parametrically upscaled constitutive damage model

2026 | 20th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM26) | Pasadena, California | Oral presentation

Virtual damage sensing digital twin for piezocomposites integrating neural operator with parametrically upscaled constitutive damage model

2026 | Enabling Multifunctionality through AI Informed Material Discovery and System Design (GRC 2026) | Ventura, California | Poster presentation

Virtual damage sensing digital twin for piezocomposites integrating neural operator with parametrically upscaled constitutive damage model

2026 | HEMI Day 2026 | Baltimore, Maryland | Podium presentation

Virtual damage sensing digital twin for piezocomposites integrating neural operator with parametrically upscaled constitutive damage model

2024 | The 16th world congress on computational mechanics (WCCM16) | Vancouver, Canada

A phase-field approach for the nucleation and propagation of dynamic cracks

2023 | The 17th united states national congress on computational mechanics (USNCCM17) | Albuquerque, New Mexico

A model-based simulation framework for coupled acoustics, elastodynamics, and damage with application to nano-pulse lithotripsy

2022 | The 15th world congress on computational mechanics (WCCM15) | Yokohama, Japan

Model-based simulations of pulsed laser ablation using a CutFEM method

2019 | The 15th united states national congress on computational mechanics (USNCCM15) | Austin, Texas

A homogenization-based phase field approach to fracture

2019 | The 2nd international conference of mechanics of advanced materials and structures | Nanjing, China

A micromechanics-based phase field approach to fracture

2019 | The international conference on data driven computing and machine learning in engineering | Shanghai, China

A manifold learning approach for multiscale phase field evolution for fracture

期刊审稿

- Mechanics of Materials
- Advances in Engineering Software
- SoftwareX